

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

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Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

You have worked with the Kisumu tray of eight substances (A table salt, B water, C sand, D sulphur, E iron filings, F antacid tablet, G paracetamol tablet, H sugar) over ten lessons. Use your own tray evidence and your learning from the lessons to answer the Driving Question: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely? Answer each section in the space under its prompt. Use the letters A-H and the exact names of the substances. Use numbers and units (g, mL, C, mg) from your own results where you can. Write in full sentences and show your working for any calculation. Remember our class value of respect: write in your own words and be ready to listen to other groups' ideas.

Section 1: Same look, different behaviour

Look at the data table from the Kisumu tray tests (1 level spatula, about 2 g, of each solid in 50 mL of water at about 25 C; heat test done by the teacher).

Substance	In water	Heat test
A table salt	dissolves, clear solution	crackles, no melting
C sand	sinks, stays visible	no change
D sulphur	floats, not wetted	melts at about 115 C, then burns with blue flame
E iron filings	sink, pulled out by a magnet	glow orange-red, turn dull blackish
H sugar	dissolves, clear solution	melts at about 186 C, browns, then chars

(a) Wanjiru noticed that several of these substances look like powders. Why do substances that look alike behave so differently? Use evidence from at least three substances in the table. (b) Name two physical properties and one chemical property that you used to tell the substances apart.

Write your answer here:

Section 2: Separating mixture M

Mixture M was 3 g iron filings + 3 g sand + 3 g salt, a total of 9 g, in water. Group 1 separated it in three steps and recorded the masses recovered: iron 2.9 g, sand 2.8 g (after drying), salt 2.7 g.

(a) Name the method that removed each of the three substances and explain why that method worked for that substance. (b) Why could the salt not be removed by filtration or a magnet? (c) Calculate the total mass recovered and the mass lost. Give one reason for the loss.

Write your answer here:

Section 3: Classifying and the meaning of Chemistry

(a) Write the meaning of Chemistry and of matter. (b) Classify D sulphur, A table salt and mixture M as an element, a compound or a mixture, and give a reason for each. (c) Name the five branches of chemistry and say which branch you would link to the separation of mixture M.

Write your answer here:

Section 4: Medicines and using them safely

Wanjiru asked why the antacid and paracetamol work differently. (a) Using what you saw in the vinegar test, explain how F antacid tablet relieves heartburn and how G paracetamol tablet is different. (b) The adult paracetamol label says 1 to 2 tablets of 500 mg every 4 to 6 hours. Calculate the maximum number of milligrams and grams allowed in 24 hours if the maximum is 8 tablets, and say what can happen if more is taken. (c) Otieno's customer wanted to give an adult dose to a small child. What advice would you give, and which items on a label help a buyer to be safe?

Write your answer here:

Section 5: Final Explanation

Write a connected explanation of 8-10 sentences that answers the Driving Question: How do the different properties of substances around us help chemists make medicines, foods and other products, and how do we use them safely? Use evidence from the Kisumu tray, including at least one number from your results. Include (i) why substances that look alike behave differently, (ii) how chemists use properties to separate or make materials, with at least one Kenyan industry or example, (iii) why the antacid and paracetamol work differently and why both are harmful in excess, and (iv) one responsibility you will take for a safe and healthy environment at school and at home.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the Kisumu tray	Uses accurate results for several substances A-H, including exact numbers and units (such as 2.9 g, 2.8 g, 2.7 g, 8.4 g, 0.6 g, 4 g), and links each piece of evidence to a claim.	Uses correct evidence from some substances and at least one correct number, but links to the claim are sometimes missing.	Gives few or inaccurate results, or makes claims without evidence from the tray.
Explaining properties and separation	Clearly explains that properties come from what a substance is made of, and correctly matches magnetic separation, filtration and evaporation to the properties of iron, sand and salt, including why dissolved salt needs evaporation.	Names the methods and most properties correctly, with a partly complete reason for why each method works.	Confuses methods or properties, or only states that substances are different without a reason.
Chemistry concepts and classification	Correctly defines chemistry and matter, classifies elements, compounds and mixtures with reasons, and names the five branches with a correct link.	Gives mostly correct definitions and classifications, with a reason missing in places.	Definitions or classifications are incomplete or mostly incorrect.
Medicines and safe use	Explains that the antacid neutralises acid and paracetamol acts differently, calculates $8 \times 500 \text{ mg} = 4000 \text{ mg} = 4 \text{ g}$	Explains the medicines and calculation correctly but gives limited detail on label items or harm.	Mixes up the two medicines, makes a calculation error, or gives little on safety.

	correctly, explains the harm of excess, and links label items, correct dosage and avoiding substance abuse to safety.		
Final explanation: clarity, Kenyan context and responsibility	Gives a connected 8-10 sentence answer to the driving question, uses a Kenyan industry or example, and states a specific, realistic responsibility for school and home.	Answer is mostly connected and answers the driving question, with a general Kenyan example or responsibility.	Answer is a list of unconnected facts, is too short, or does not address the driving question.

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